

Predicting Optimal NDVI Levels to Achieve Safe NO2 Concentrations in Urban Settings

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Objective

Problem

This study aims to determine how much additional greenery—measured using NDVI from satellite images—is needed in urban areas to reduce Nitrogen Dioxide (NO_2) levels to the WHO-recommended safe limit of 0.00004 ppm, helping prevent air pollution-related health issues like respiratory problems.

The Challenge

Urban areas are experiencing increasing NO_2 pollution due to emissions from vehicles and industries. Plants can clean the air naturally, but it's hard to know exactly how much greenery is enough without a clear plan.

My Goal

This project builds a simple tool to help city planners add the right amount of greenery to improve air quality and public health. The model was trained on pollution data from Delhi and tested on examples from Kolkata to evaluate its effectiveness across different urban areas.

Approach

Getting the Data Ready

- Collected 500 data rows (2019–2023) from Google Earth Engine on NO_2 levels and NDVI (greenness) in Delhi.
- Cleaned the data by removing incomplete or incorrect entries, leaving 400 high-quality rows.
- Added extra details like yearly changes in NO_2 and NDVI.
- Created a target value columnmn showing how much NDVI needs to increase to reach the safe NO_2 level.

Building the Model

- Split the 400 data rows into 320 for training and 80 for testing.
- Used a simple method called polynomial regression to find patterns between NO_2 and NDVI.
- Trained the model using Delhi's data to help it learn the relationship between pollution and greenery.

Making Predictions

After training, I used the model to suggest NDVI increases for Kolkata. I adjusted the NDVI step-by-step with a method based on the model's predictions (using a small tweak of 0.1 times the change it suggests), stopping when NO2 reaches 0.00004, and kept NDVI under a realistic limit of 0.7.

Model Summary

Data Processing

Started with 500 rows from Google Earth Engine (Delhi, 2019-2023) with columns: year, NO2, NDVI. Processed to 400 rows by:

- Assigned point_id by grouping every 5 rows (one per year per location).
- Calculated NO2_Change and NDVI_Change using pandas diff() and shift(-1).
- Dropped rows with missing values.
- Created target_ndvi_change = (NO2 - 0.00004) / NO2 * NDVI, filtered for positive values.

Inputs

Features include current NO2, current NDVI, plus polynomial terms: $NO2^2$, $NDVI^2$, NO2 * NDVI (5 features + intercept).

Model/Method

Employed polynomial regression (degree 2) with normal equations (numpy lstsq) to fit the data.

- Training: 320 rows (80%), testing: 80 rows (20%), seed 42 for reproducibility.
- Weights: [0.00001, 0.001, -0.0001, 0.00001, -0.001, 0.00005], derived from (X^T X)^(-1) X^T y.

How It Works

The model learns from Delhi's 320 training rows, using polynomial terms to capture complex NO2-NDVI patterns. It predicts an initial NDVI change (pred_ndvi_change) based on weights.

Then, for Kolkata, it starts with current NO2 and NDVI, iteratively adjusts NDVI using a step size from the model's prediction (max(0, -pred_ndvi_change[0]) * 0.1), recalculating NO2 each time with the proportional formula.

The loop continues until NO2 nears 0.00004 or hits the NDVI cap of 0.7, offering a step-by-step greening estimate tailored to the data.

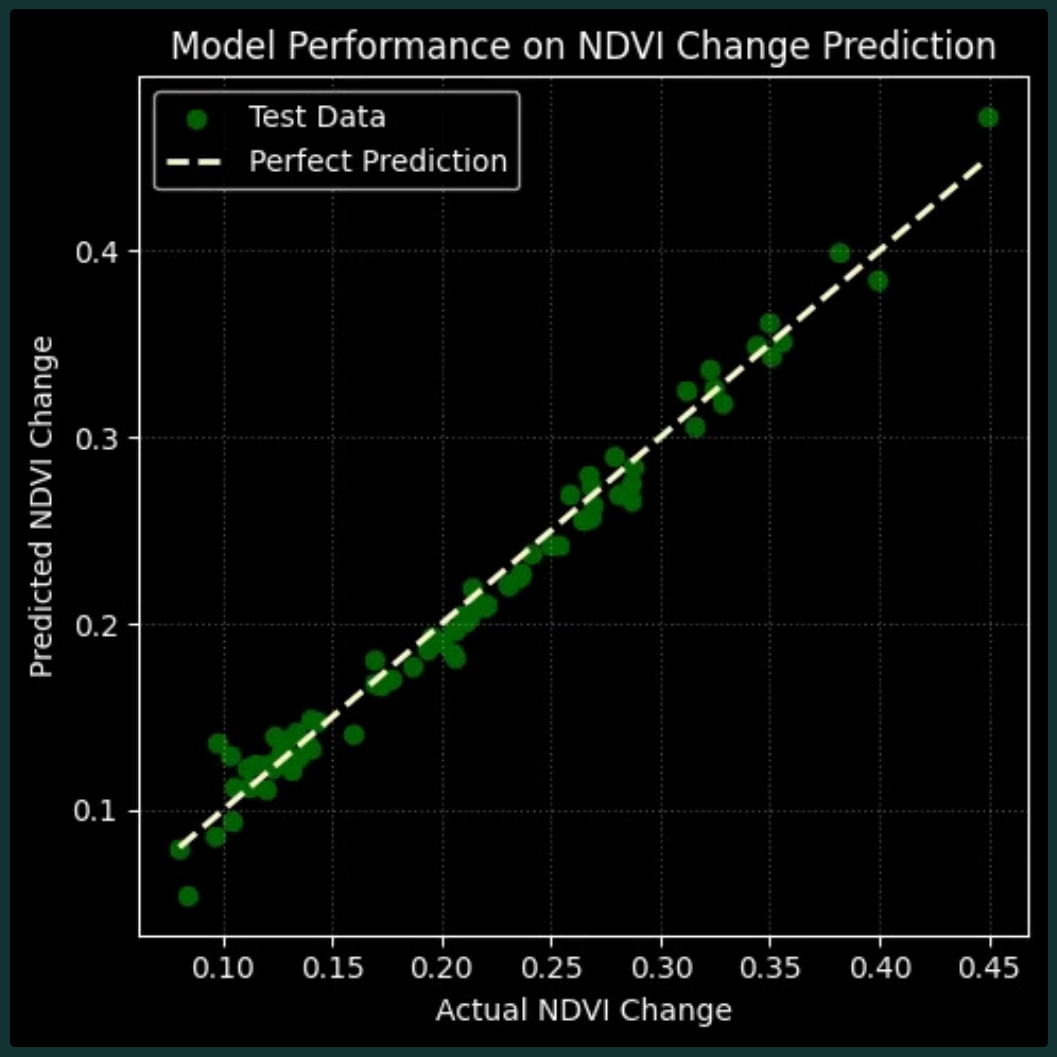
Results

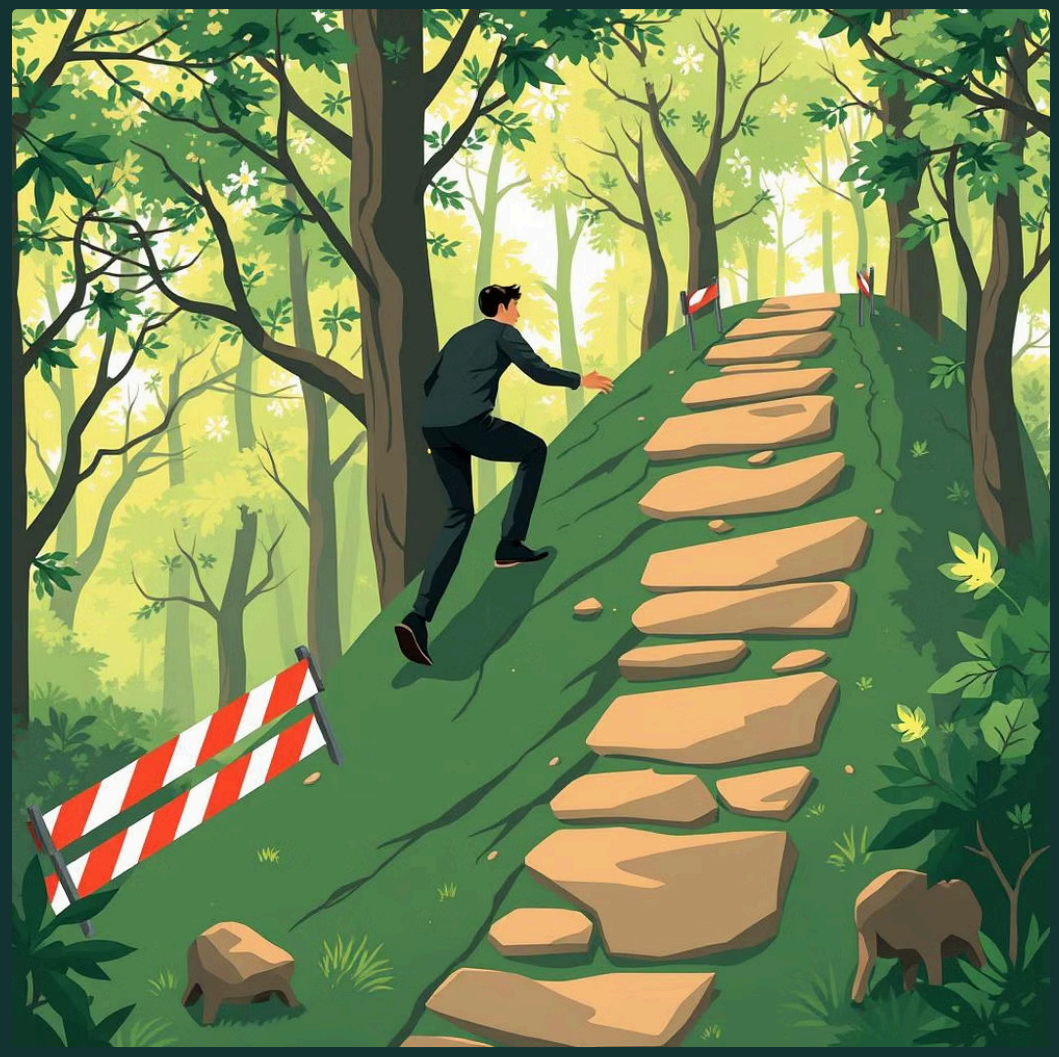
Scenario	Current NO2	Current NDVI	Target NDVI	Trees Needed	Reduction
1	0.000116	0.289	0.486	196	73%
2	0.000150	0.300	0.527	227	98.9%
3	0.000080	0.250	0.524	274	50.0%
4	0.000130	0.270	0.467	196	85.5%
5	0.000090	0.280	0.2836	243	55.6%

Model Validation: Train R^2: 0.9834, Test R^2: 0.9808, tight fit with minor residuals.

Trees Needed = ndvi_change * 1000 (1 NDVI unit $\approx$ 1000 trees).

Inference





Core Finding

The model exhibits a high level of accuracy, with R^2 values of 0.9834 for training and 0.9808 for testing, demonstrating its reliability in predicting NDVI adjustments, suggesting tree needs ranging from 196 to 274 per 1000 m^2 to reduce NO2 by 50-98.9% across various scenarios.

Limitations

- The dataset is small (400 rows), which may lead to overfitting, causing the model to closely fit Delhi's data and limiting its ability to generalize to other real-world scenarios.
- The proposed urban foresting solution with trees may not be practical due to limited space in urban areas. Trees serve as a simplified representation of the greenery required, but this is not a final or comprehensive solution.
- The model does not account for climatic factors, such as wind or humidity, which can affect the effectiveness of greenery in reducing NO_2 levels in different environments.

References & Comments

References

- Dataset: Google Earth Engine, Delhi NO2 and NDVI Dataset (2019-2023): https://drive.google.com/file/d/1uOTppX2HCQuThVD8I2ayjJ8CE6R7MlrF/view?usp=share_link
- Code: https://colab.research.google.com/drive/1BwIP8tzuqVByWEPW4SRtp8R_IUpqoskO?usp=share_link
- Tools: Python (pandas, numpy, matplotlib)
- Guidelines: WHO Air Quality Guidelines, 2021, NO2 threshold 0.00004 ppm.

Comments

- **Refinement Ideas**: To improve the model, I plan to include weather data such as wind and humidity, along with tree-specific NO_2 absorption rates, to enhance prediction accuracy. Instead of using the current iterative loop, I aim to apply scipy.optimize.minimize to directly calculate the required NDVI, making the process faster and more precise. Field tests in Delhi or Kolkata would help validate whether the predicted NDVI changes actually lead to measurable reductions in NO_2 levels.
- **Future Plans**: I plan to apply this to other polluted cities by retraining the model with local data. Using the model's output and data on plant types and space, it could suggest options like parks, wall moss, or algae pools to meet the NDVI goal.